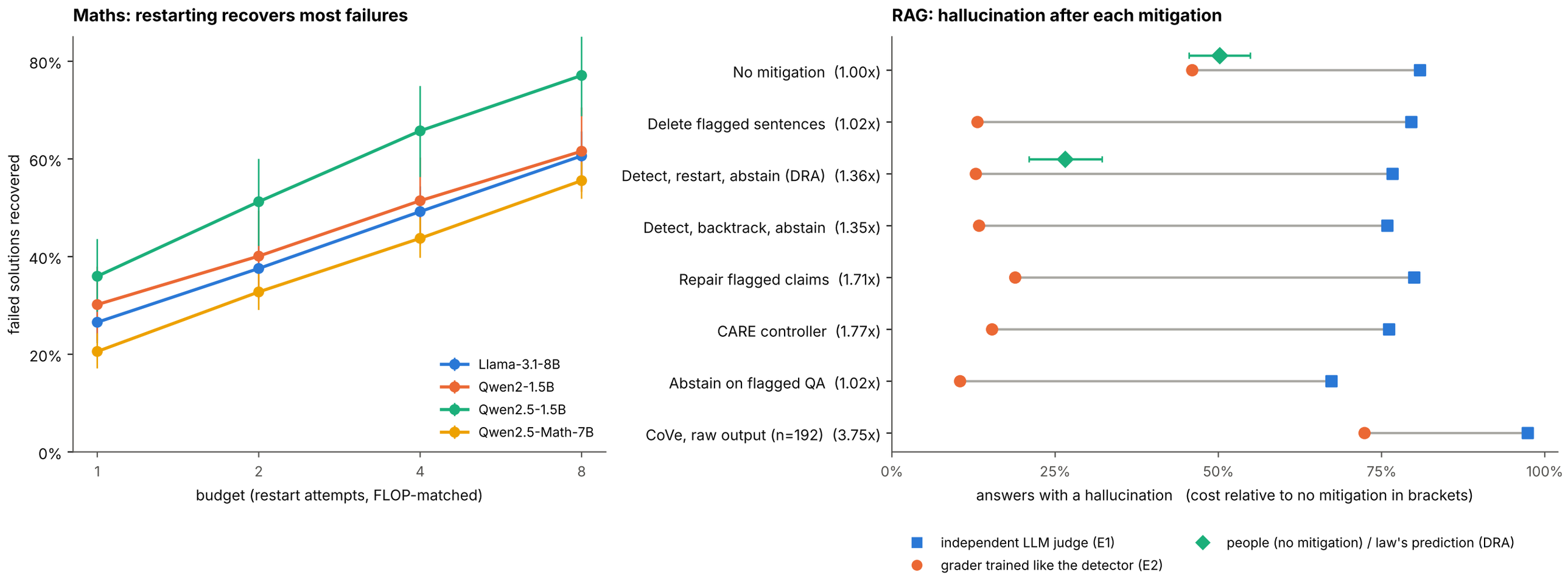
Maths: restarting recovers most failures
failed solutions recovered
80%
60%
40%
20%
0%
1
2
4
8
budget (restart attempts, FLOP-matched)
Llama-3.1-8B
Qwen2-1.5B
Qwen2.5-1.5B
Qwen2.5-Math-7B
RAG: hallucination after each mitigation
No mitigation (1.00x)
Delete flagged sentences (1.02x)
Detect, restart, abstain (DRA) (1.36x)
Detect, backtrack, abstain (1.35x)
Repair flagged claims (1.71x)
CARE controller (1.77x)
Abstain on flagged QA (1.02x)
CoVe, raw output (n=192) (3.75x)
0%
25%
50%
75%
100%
answers with a hallucination (cost relative to no mitigation in brackets)
independent LLM judge (E1)
grader trained like the detector (E2)
people (no mitigation) / law's prediction (DRA)